Chapter 13, Problem 72P

Problem

In order to meet an emergency, three single-phase transformers with 12,470/7,200 V rms are connected in Δ-Y to form a three-phase transformer which is fed by a 12,470-V transmission line. If the transformer supplies 60 MVA to a load, find:

(a) the turns ratio for each transformer,

(b) the currents in the primary and secondary windings of the transformer,

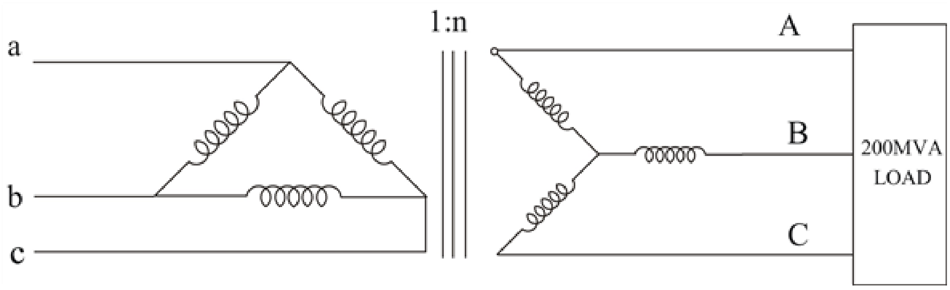
(c) the incoming and outgoing transmission line currents.

Step-by-step solution

Step 1 of 5

(A) Consider just one phase at a time,

Step 2 of 5



Step 3 of 5

$$n = \frac{V_L}{\sqrt{3}V_{LP}} = \frac{7200}{(12470\sqrt{3})} = \frac{1}{3}.$$

Step 4 of 5

(B) The load carried by each transformer is,

$$\frac{60}{3} = 20\ MVA$$

Hence,

$$I_{LP} = \frac{20\ MVA}{12.47\,K} = 1604\ A$$

$$I_{LS} = \frac{20\ MVA}{7.2\,K} = 2778\ A.$$

Step 5 of 5

(C) The current is incoming line is A, B, C is,

$$\sqrt{3}\ LP \quad = \quad \sqrt{3} \times 1603.85 \quad = \quad 2778\ A$$

Current in each outgoing line A, B, C is,

$$\frac{2778}{\left(n\sqrt{3}\right)} \quad = \quad 4812\ A\ .$$